

WEB PROGRAMMING LAB-10 HOME EXERCISES

📅 Date April 1, 2026

Name: Devadathan N R

Registration No: 230905010

Class: CSE - A1

Roll No: 04

Title: Databases – Part I

Question 1

Assume a table "Institutes" with institute_id, name, and no_of_courses are the fields. Create a web page that retrieves all the data from "Institutes" table displays only Institute names in the list box.

Create project & App

```
django-admin startproject instituteproject
cd instituteproject
python manage.py startapp instituteapp
```

`instituteproject/settings.py`

```
import os # added at the top

INSTALLED_APPS = [
    'instituteapp',
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
]

STATIC_URL = '/static/'
STATICFILES_DIRS = [BASE_DIR / 'static']
```

Project URLs: `instituteproject/urls.py`

```
from django.contrib import admin
from django.urls import path, include
```

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('instituteapp.urls')),
]
```

App URLs: `instituteapp/urls.py`

```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path('add/', views.add_institute, name='add_institute'),
]
```

Models: `instituteapp/models.py`

```
from django.db import models

class Institute(models.Model):
    institute_id = models.CharField(max_length=20, primary_key=True)
    name = models.CharField(max_length=200)
    no_of_courses = models.IntegerField()

    def __str__(self):
        return self.name
```

Forms: `instituteapp/forms.py`

```
from django import forms
from .models import Institute

class InstituteForm(forms.ModelForm):
    class Meta:
        model = Institute
        fields = ['institute_id', 'name', 'no_of_courses']
```

View: `instituteapp/views.py`

```
from django.shortcuts import render, redirect
from .models import Institute
from .forms import InstituteForm

# View to display the list box
def home(request):
```

```

institutes = Institute.objects.all()
return render(request, 'home.html', {'institutes': institutes})

# View to populate the database manually
def add_institute(request):
    if request.method == 'POST':
        form = InstituteForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('home')
    else:
        form = InstituteForm()
    return render(request, 'add_institute.html', {'form': form})

```

Templates

home.html

```

<html>
<body>
    <h2>Institute Directory</h2>
    <a href="/add/">Add New Institute</a>
    <hr>
    <h3>Select Institute:</h3>
    <select name="institute_listbox" size="5" style="width: 250px;">
        {% for inst in institutes %}
            <option>{{ inst.name }}</option>
        {% empty %}
            <option>No data available</option>
        {% endfor %}
    </select>
</body>
</html>

```

add_institute.html

```

<html>
<body>
    <h2>Add Institute Data</h2>
    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Save Institute</button>
    </form>
    <br>
    <a href="/">Back to Directory</a>
</body>
</html>

```

Run

```
python manage.py makemigrations
python manage.py migrate
python manage.py runserver 8080
```

Output

Institute Directory

[Add New Institute](#)

Select Institute:

MIT
KMC

Add Institute Data

Institute id:

Name:

No of courses:

[Back to Directory](#)